Systems Administration (ASIST)

Practical Theoretical Class 07

Active Directory Users and groups
Active Directory policies
Active Directory trusts

Active Directory - LDAP

The logical structure of Active Directory domains (*domains*, *domain trees*, *forests*) is supported by databases residing in each domain (replicated between the various DCs of the same domain).

The directory service used to access these databases is called *Active Directory* (**AD**), which is based on the LDAP protocol. Each AD domain is identified by a DNS name, corresponding to the domain DIT **root-DN**. As an LDAP DIT, the domain is organized into branches.

Objects such as users and groups are housed in DIT branches, some branches are system-defined, such as **Builtin**, **Computers** and **Users** (they are *Container* class). Administrator can define new branches by creating objects of ***Organizational Unit*** class.

Many of the *Active Directory* management tools protect users from interacting directly with LDAP, including by not identifying objects by *distinguished name*(**DN**).

In an AD domain, two objects can have the same **RDN** as long as the **DN** is different, that is, two objects can have the same name as long as they are in different branches.

From an administration point of view, dividing the domain into branches has several advantages, but the need to refer to the location of each object within the domain turns out to be a drawback.

AD – user and group identification

Pre-AD Windows domains from an object naming perspective were completely flat.

Users and groups are identified as **DOMAINNAME\USERNAME** and **DOMAINNAME\GROUPNAME,** where *DOMAINNAME* is the NetBIOS domain name (unqualified AD domain name).

This form is still supported in AD domain, both for groups and users in the case of users is even way more used to identify the login (***Down-level Logon Name***).

However, this way of identifying users and groups is not sufficient in AD because within the domain there may be multiple groups and users with the same name.

To resolve this issue in AD each group account has a **pre-Windows 2000** name to be used in the form **DOMAINNAME\groupname.** This name is usually the same as the group account name, however it must be unique throughout the domain.

The same is true for user accounts, the **Pre-Windows 2000** name is commonly used when logging in as **DOMAIN\username** (*Down-Level Logon Name*), defaults to the account name. Two users can have accounts with the same name in their domain AD, but their *Down-Level Logon Name* must be different.

AD – *User principal name* (UPN)

The most current way to identify a user in AD is in the UPN form, also called *User Logon Name* because it is used at logon:

username@AD-DNS-DOMAIN

By default the UPN is set to the user account name (object RDN) and the DNS name of the AD domain. In a domain, two UPNs cannot be the same, if there are two user accounts with the same name it is necessary for one of them to use an UPN different than the account name.

A user account in the domain is an LDAP object identified by DN, or a **cn** (*Common Name*) attribute is used in the RDN, or its value is an account name, for example **cn=specialuser**.

In this example the UPN will be **specialuser@AD-DNS-DOMAIN** and the name Pre-Windows 2000 will be **DOMAIN\specialuser**. These will be the default values, can be changed by the administrator if necessary or convenient.

Also group accounts are identified by the **cn** (*common name*) attribute corresponding to the account name. Therefore, a user account and a group account can have the same name only if they have different branches, but Pre-Windows 2000 names must necessarily be different.

AD – groups

Group accounts in an AD domain are quite different from **local group** accounts that only exist on the local machine. In one domain group accounts are stored in the domain's AD base and are valid throughout the domain.

In AD groups are LDAP objects identified by a DN, within the domain database (DIT) can be located in any branches, the **Pre-Windows 2000 name** attribute (*DOMAIN\group*) allows to identify the group in the domain without the need to identify the branch. Contrary to what happens with a local group, AD group might have as members other groups (group *nesting*).

Machines that are members of a domain can continue to work with local users and groups, however domain controllers do not have local accounts.

There are two group types in AD:

Security group – are security objects, can be used to assign rights to their members, and can be included in ACLs.

Distribution group – are intended to create logical groups of users and groups for other purposes. Does not allow assign rights to members nor can they be used in ACLs. They can be used for example to create email distribution lists. Note that a *security group* can also be used to distribute mail.

AD – *group scope*

Group scope defines its coverage: in terms of members who may have, groups that can be a member, and permissions or rights that may be associated.

Local groups (Local scope) – exist only locally on the machine (not in AD). Can include forest-wide members and other trusted domains(*trusted domains*). The rights assigned to the group can only be related to the local resources of the machine where they are set. A local group cannot be a member of other groups.

Domain local groups (Domain local scope) – exist in the AD domain. May include as **members:** users and global groups from across the forest and other trusted domains (*trusted domains*), universal groups of the forest domain and local groups from the same domain. **Can be members of:** *local domain* groups from the same domain and local machine groups from the same domain (excluding built-in groups). Rights assigned to the group may relate to any object or resource in the domain, but not resources from other domains.

AD – *Global and Universal groups*

Global groups (Global scope) – exist in the AD domain. **Possile members:** users and global groups from the same domain. However a *global group* can be **member of** universal groups from the same forest, global groups from the same domain and local groups defined on trusted domain machines. It is possible to assign to a *global group* permissions on any domain in the forest or trusted ones.

Universal groups (Universal scope) – exist in the AD domain. **Possile members:** forest users, global and universal groups of the same forest. **Can be members of:** universal groups of the forest, *domain local* groups from trusted domains (includes all domains in the forest) and local groups defined on trusted domain machines. Group rights can relate to any existing object or resource in the forest and other *trusted domains*.

A **universal** group can be converted to ***domain local*** or **global**.

A **global** group can be converted to **universal**.

A ***domain local*** group can be converted to **universal**.

These conversions are only valid if they do not violate the previously stated rules.

Organizational Units (OUs)

In the context of an AD domain objects such as users, computers and groups are in LDAP branches. Although some branches such as **Builtin**, **Users**, and **Computers** that are ***Container*** are initially defined in AD, new branches can be defined through *Organizational Unit* (**OU**) objects.

OUs allow to subdivide a domain into logical areas in a tree structure (one OU can contain other OUs). Objects such as users and groups can be moved from one OU to another OU within the same domain.

There are several advantages to defining OUs. From an organizational point of view, it becomes more apparent to administrators that affinity relationships between subsets of users and groups are organized in OUs. Initially, it is defined the **Domain Controllers** OU that contains objects of the type **Computer** corresponding to the various domain DCs.

OUs allow:

- **Delegate powers of administration** – for a given OU the administrator can delegate their management to certain users and groups, defining exactly what powers are delegated.
- **Set group policies** (**GPO** – *Group Policy Objects*) to apply to the OU and therefore to all objects that are in the OU, such as user accounts.

Appointment of group managers and OUs

In the case of an OU, the recommended way to assign management powers to users or groups over the OU and objects residing in it is through delegation (**Delegate Control**).

However objects in the **Group**, **Computer**, and **Organizational Unit** classes have the **managedBy** attribute that identifies the user or group that is responsible for managing them. This is an alternative way of assigning administration rights to an OU, which however seems to present some problems in specific situations.

Policies

A policy is a **set of configuration elements**, including user-related and computer-related configuration elements. Configuration elements can be as diverse as **Start Menu** and **Desktop** entries, **registry** keys (of user and of computer), firewall settings, and permissions (**user rights**) to assign to the user.

A configuration element may or may not be defined in a policy:

- If this is set, it is applied to the user/computer by **deleting previous settings** (original, or resulting from previous applications of other policies).
- If not set does not affect the current value of this element.

Policy Enforcement - Local Policy

Policies apply when the user logs in (when he become a member of the **Authenticated Users** group).

However, policies affect not only the user but also the configuration of the computer where he logs in.

In the case of a local login using a local computer account, there is a single applied policy called **Local Policy** that is stored locally on the computer and can be managed with the **Local Security Policy** tool.

In the case of a domain login, the **Local Policy** defined on the computer used is applied first, but then the application of policies defined in *Active Directory* that will override Local Policy.

In the context of an *Active Directory* domain policies are stored in *Group Policy Objects* (**GPO**) there are some GPOs defined but others can be created.

In a domain GPOs can be managed through the **Group Policy Management** tool. A GPO is a set of configuration elements to be applied during login of users is necessary to establish appropriate connections between the object field and the GPO.

GPO (*Group Policy Objects*) application

When a user logs in to the domain, Local Policies are initially applied, following are the policies defined in *Active Directory* GPOs.

Once GPOs are defined, three types of AD objects can be linked to them: **Site**, **Domain** and **OUs**.

Different Sites, Domains or OUs may be linked to the same GPO, also a Site, Domain or OU may be linked to several GPOs.

The order of application of GPOs is notoriously relevant because a GPO overlaps with previously applied GPOs (only for the configuration elements it defines).

Application order follows *Active Directory* hierarchy: first the GPOs to which the **Site** is linked are applied, then the GPOs to which the **domain** is linked and finally the GPOs to which **each OU** is linked (following the hierarchy between different OUs).

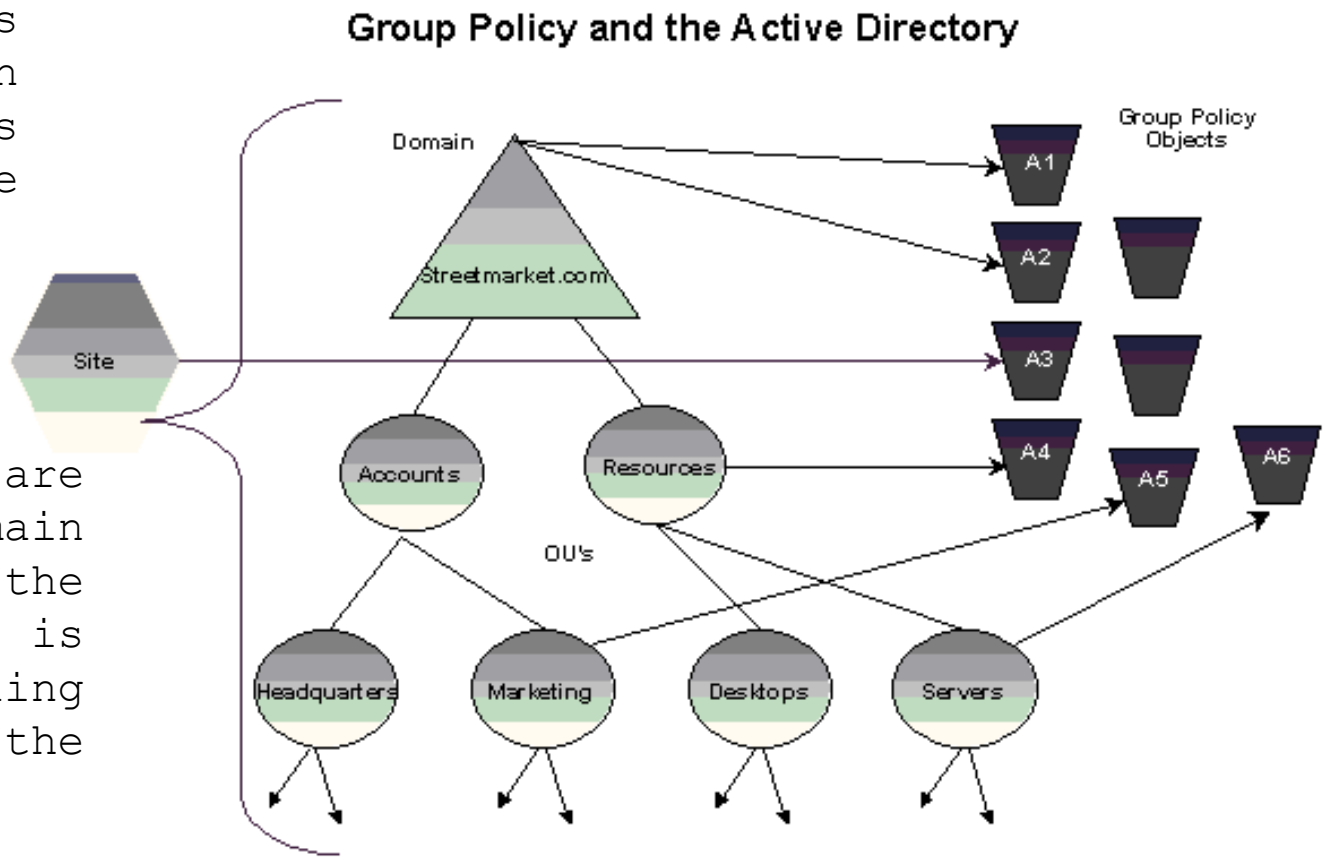
When a site, domain or OU is linked to multiple GPOs we can define exactly the order of application of the various GPOs (***link order***).

GPO (*Group Policy Objects*) application

The figure below (*https://technet.microsoft.com*) shows an example of policy enforcement in *Active Directory*.

Once *Local Policy* is applied (not shown in the image) GPO A3 is applied because the Site is linked to it..

Next, GPOs A1 and A2 are applied because the domain is linked to them, the order of application is established by defining domain connections to the GPOs (*link order*).



GPO (*Group Policy Objects*) application

GPOs are applied to **User** and **Computer** objects, not **Group** objects. In other words, GPOs applied to a user are established only by the user account (namely the OU in which it is defined) and **not by the groups of which it is a member.**

Specifically, if a user is a member of a group and that group is defined in an OU that is linked to a GPO, then that is no reason why that GPO will be applied to the user during login.

A different scenario is where the user is defined in an OU linked to a GPO and that GPO gives a group certain permissions (*user rights*). In this case the GPO will be applied to the user during login. Consequently, if the user is a member of that group, they will have the permissions that have been assigned to the group.

Active Directory Trust Relationships

Cross-domain trust relationships can be established **two-way trust** or only **one-way trust**.

A *one-way trust* can be used when we are interested in one domain being able to access the resources of another domain, but not the other way around.

Trusts can be transitive (***transitive trust***) or nontransitive (***nontransitive trust***). Being transitive means that trust extends across domains, for example: if domain A trusts domain B (A-->B) and domain B trusts domain C (B-->C) then domain A trusts domain C (A-->C):

A --> B --> C => A --> C

In a forest there is implicitly a ***two-way transitive trust*** between a domain and the hierarchically superior domain (parent). Because the trust is transitive, it extends to all domains of the forest, so between any two domains in the same forest there is always a ***two-way trust***.

In addition to trusts thus established between domains in the same forest we can define other trusts, provided that we have the necessary administration permissions on the domains and forests involved.

Active Directory – External Trust and Forest Trust

An ***external trust*** can be established between one domain in one forest and another domain in another forest. It may be **one-way** or **two-way**, but it is ***nontransitive***. Because it is non-transitive, it does not spread to other forest areas and is confined to the two areas between which it is established. According to the directions in which it is established, it allows users in one domain to access resources in the other domain.

A ***forest trust*** is somewhat equivalent, but at the forest level. It can be **one-way** or **two-way**, but it is always ***transitive***. When established it applies to the entire forest, ie all domains of the forest. According to the sense in which it is established, it allows users from any domain in one forest to access resources from any domain in another forest.

Active Directory – Realm Trust and Shortcut Trust

A ***realm trust*** is used to establish a trust relationship between a *Windows Active Directory* domain and a *Kerberos version 5 realm* from non-Windows systems, for example Linux.

Realm trust can be **one-way** or **two-way** and can be ***transitive*** or ***nontransitive***.

A ***shortcut trust*** is always ***transitive***, but it can be **one-way** or **two-way**. This trust relationship can be defined within a forest to improve the performance of operations between different trees of the same forest. Validation of the trust relationship between two distinct tree domains is established using trust relationships through the AD hierarchy down to the forest level.

From an efficiency standpoint this could be penalizing, for example when a user from one domain in one tree logs in to a domain in another tree. ***Shortcut trusts*** allows to avoid going to the forest level for this type of operation.